

Lecture 14 Complex Bollobás-Erdős graphs.

Recall

$$2 \leq p < q$$

$$RT_p(n, K_q, \epsilon n) = \max \left\{ e(G) : \begin{array}{l} |G| = n; \\ K_q\text{-free}; \\ \alpha_p(G) \leq \epsilon n \end{array} \right\}$$

where $\alpha_p(G) = \max |U| : U \subseteq V(G) \text{ s.t. } G[U] \text{ is } K_p\text{-free}$

$$P_p(K_q) = \lim_{\epsilon \rightarrow 0} \lim_{n \rightarrow \infty} \frac{RT_p(n, K_q, \epsilon n)}{\binom{n}{2}} \quad \text{Ramsay-Turán density.}$$

Szemerédi

$$P_2(K_4) \leq \frac{1}{4} \quad \left(\forall n\text{-vx graph } G : K_4\text{-free}, \alpha(G) = o(n) \Rightarrow e(G) \leq \left(\frac{1}{4} + o(1) \right) \binom{n}{2} \right)$$

$$\bullet \text{ Bollobás-Erdős } P_2(K_4) \geq \frac{1}{4}, \text{ i.e.}$$

$$\exists n\text{-vx graph } G \text{ s.t. } - e(G) \geq \left(\frac{1}{8} - o(1) \right) n^2$$

$$- K_4\text{-free}$$

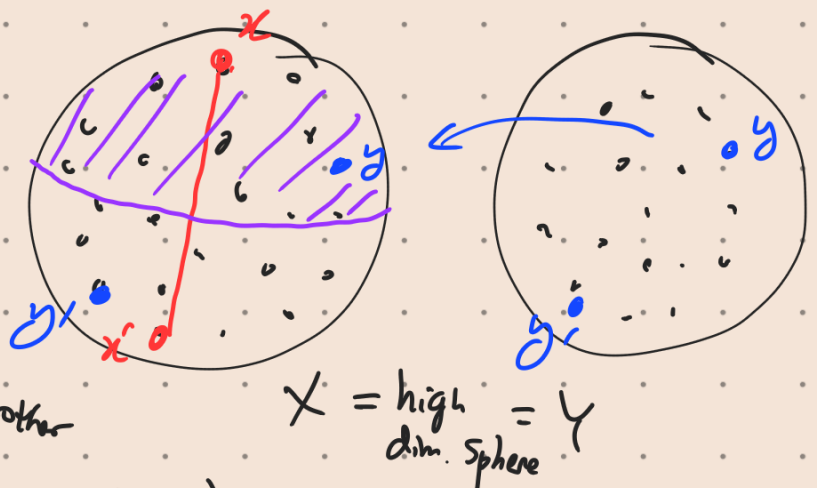
$$- \alpha(G) = o(n).$$

Construction:

Inner: $x \sim x'$
 $(\Rightarrow)$ almost antipodal

Cross edges: $x \sim y$
 $(\Rightarrow)$ if one is in the hemisphere centered at the other

(in the pic. $x \sim y, x \not\sim y'$)



$$\Rightarrow \text{every } vx \sim \text{half of } vs \text{ from the other side} \\ \Rightarrow e(G) \geq \frac{n^2}{8} + o(n^2)$$

- $\alpha(G) = o(n)$

- $\forall X \subseteq V(G), |X| = \Omega(n)$

- $\Rightarrow G[X]$ has an edge



Isoperimetric ineq.

Given measure, spherical cap minimises diameter

$$\text{diam}(X) \geq \text{diam}(C), \quad \lambda(X) = \lambda(C) = \Omega(n)$$

upp. bd. on measure of sph. cap

$\Rightarrow \exists$ two pts almost antipodal in $X \Rightarrow$ form an edge.

- BE graph is K_4 -free

Details: • k dim. of high dim. sphere

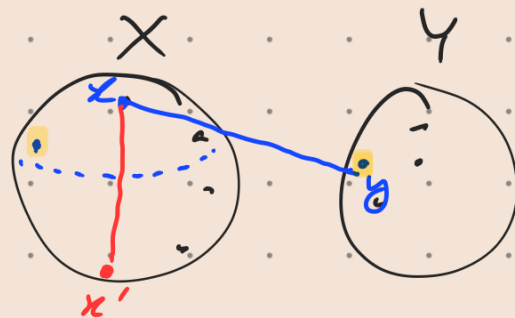
- $\mu = \frac{\varepsilon}{\sqrt{k}}$

Lem.: $\exists C > 0$ st. TP.H. $\forall n \geq \left(\frac{C}{\delta}\right)^k$, can cut

S^{k-1} into n pieces of the same measure, each w./ diameter $\leq \delta$.

- $x \sim y \Leftrightarrow |x - y| < \sqrt{2} \mu$

- $x \sim x' \Leftrightarrow |x - x'| > 2 - \mu$



Claim X, Y are Δ -free (high odd girth)

PF: Supp. $x_1, x_2, x_3 = \Delta$



$$3 \cdot (2-\mu)^2 \leq \sum_{i,j \in \binom{[3]}{2}} |x_i - x_j|^2 = 2 \sum_{i \in [3]} |x_i|^2 - 2 \sum_{i,j \in \binom{[3]}{2}} \langle x_i, x_j \rangle$$

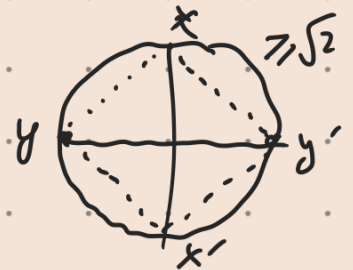
$$= 3 \sum_{i \in [3]} |x_i|^2 - |x_1 + x_2 + x_3|^2$$

$$\leq 3 \sum_{i \in [3]} |x_i|^2 = 3 \cdot 3 = 9$$



Claim G : K_4 -free

Pf: Supp. $x, x' \in X, y, y' \in Y = K_4$.



$$4(2-2\sqrt{2}\mu) = 4(\sqrt{2}-\mu)^2 > |x-y|^2 + |x-y'|^2 + |x'-y|^2 + |x'-y'|^2$$

$$8 - 8\sqrt{2}\mu + O(\mu^2) = |x+x-y-y'|^2 + |x-x'|^2 + |y-y'|^2$$

$$> 2 \cdot (2-\mu)^2 = 2(4-2\mu+\mu^2) = 8-4\mu + O(\mu^2)$$



Two main problems (sos conj)

i) $P_p(K_q)$ periodic.

ii) Construct B-E type graph w/ other densities.

$p=2$	$q=4$	$q=5$
	$q=6$	$q=7$

$$P_3(K_5) = ?$$

$$\frac{1}{8} \leq P_3(K_5) \leq \frac{1}{6} \binom{n}{2} = \frac{n^2}{12} + o(n^2)$$

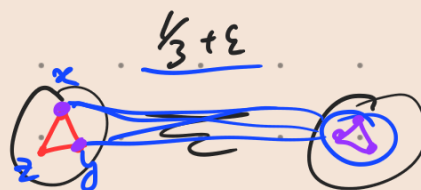
Thm (upp bd) $\forall$ n -vk graph w/ $\alpha_3(G) = o(n)$

$$\& K_5\text{-free} \Rightarrow e(G) \leq \frac{n^2}{12} + o(n^2)$$

PF: $G \xrightarrow{\text{Reg Lem}} R$ (reduced graph)

NTS $\left\{ \begin{array}{l} R \text{ is } \Delta\text{-free} \\ \text{edge weight in } R \leq \frac{1}{3} \end{array} \right.$

$$\frac{n^2}{12} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{n^2}{12}$$



Complex Bollobás - Erdős graphs (CBE)

(based on L. - Reiter - Sharifzadeh - Staden)

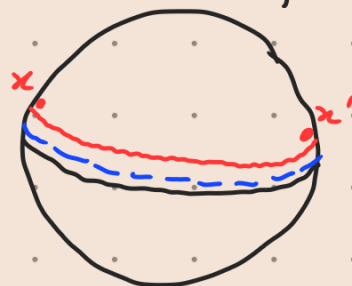
Fix h dimension ($h \rightarrow \infty$)

$$S^{h-1}(\mathbb{C}) = \left\{ (z_1, \dots, z_h) \in \mathbb{C}^h : \sum |z_i|^2 = 1 \right\}$$

$$\uparrow$$

$$\langle x, y \rangle = \sum x_i \cdot \bar{y}_i$$

$$|x - y|^2 = |x|^2 + |y|^2 - 2 \operatorname{Re}(\langle x, y \rangle)$$

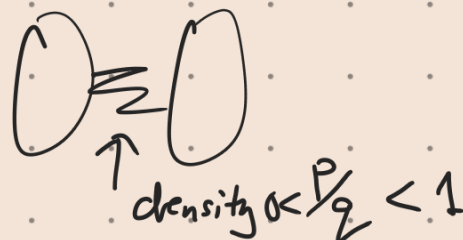


• isometry $(x_1 + y_1 i, x_2 + y_2 i, \dots, x_h + y_h i)$

$$| \longrightarrow (x_1, y_1, x_2, y_2, \dots, x_h, y_h)$$

distance-preserving map.

Construction (CBE)



$$1 \leq p < q$$

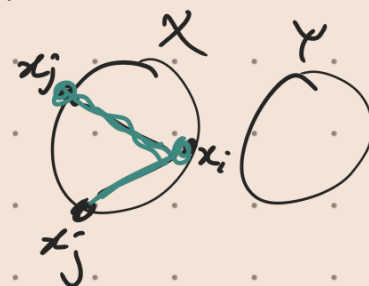
$$\text{Set } w = \cos \frac{2\pi}{q} + i \sin \frac{2\pi}{q} = e^{\frac{2\pi i}{q}}$$

$$\varepsilon \ll K^{-1} \ll q^{-1}, \quad h \rightarrow \infty$$

$$\mu = \frac{\varepsilon}{\sqrt{2h}}, \quad n \geq \left(\frac{C}{\mu}\right)^{2h}$$

Partition $\mathcal{S}^{h-1}(\mathbb{C})$ into $n/2$ equal pieces of same measure & diam $< \mu/4$.

Say $x_1, \dots, x_{n/2}, y_1, \dots, y_{n/2}$.



$$G = (V, E) \quad V = X \cup Y$$

$$x_i x_j \in E \Leftrightarrow \exists k \in [q-1] \text{ s.t.}$$

$$|x_i - \omega^k x_j| < \sqrt{\mu} \quad (\text{same for } Y)$$

$$x_i y_j \in E \Leftrightarrow \begin{cases} \text{a) } \forall k \in [q] \Rightarrow |\operatorname{Im}(\omega^k (x_i - y_j))| \geq \sqrt{\mu} \\ \text{b) } \exists \alpha \in [0, \frac{2\pi p}{q}] \text{ s.t.} \end{cases}$$

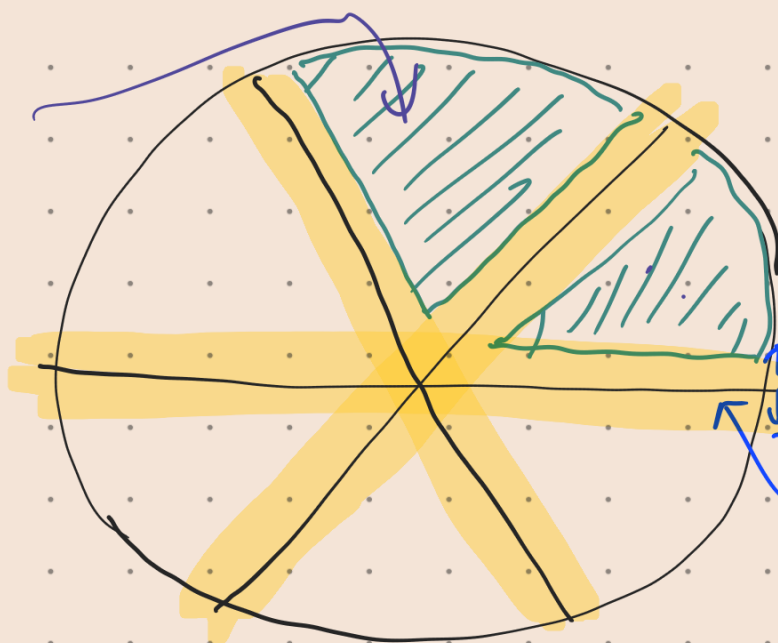
$$e^{-i\alpha} (x_i - y_j) \in [0, 1]$$

Picture

when $p=1, q=3$

$(x_i - y_j)$

(b)



yellow strips are excluded part by (a)

(b) $\Rightarrow$ every $u \sim \frac{p}{q}$ -fraction of u s from the other side

$\Rightarrow p/q$ cross density.

$$p=1 \quad q=3 \Rightarrow \frac{1}{3} \cdot \frac{n}{2} \frac{n}{2} = \frac{n^2}{12} \text{ edges}$$

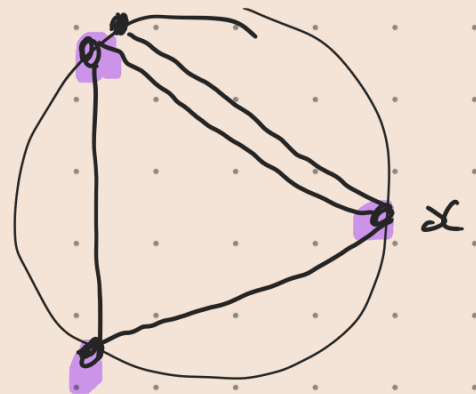
$$\Rightarrow p_3(K_5) = \frac{1}{6}$$

$$p=1 \quad q=3$$

Claim: X, Y is K_4 -free.

Claim $\alpha_3(G[X])$

$$\alpha_3(G[Y]) = o(n)$$



Follows from isoperimetric inequality & concentration of measure.

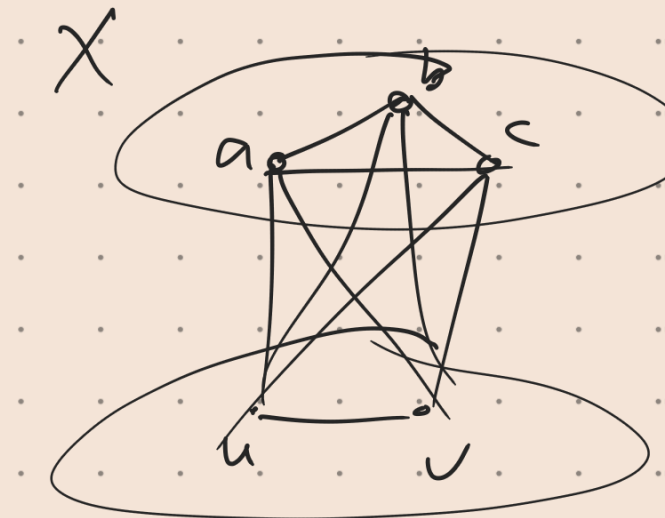
Claim K_5 -free

Inner edges

$$\Rightarrow |a - b \cdot c| < \sqrt{\mu}$$

$$|a - c \cdot w| < \sqrt{\mu}$$

$$|u - v \cdot w| < \sqrt{\mu}$$

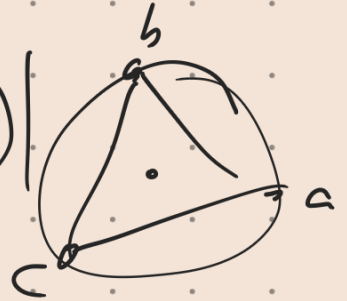


where

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad 1 + \omega + \omega^2 = 0$$

$$\Rightarrow a + \underline{a \cdot \omega} + a \cdot \omega^2 = 0$$

$$|\underline{a+b+c}| = |(b-a\omega^2) + (c-a\omega)|$$



$$\leq |b-a\omega^2| + |c-a\omega|$$

$$= |\cancel{\omega^2}(b\omega - a)| + |\cancel{\omega}(c\omega^2 - a)|$$

$$\leq 2\sqrt{\mu}$$

$$\left| \frac{a+b+c}{3} \cdot (u-v\omega) \right| \leq \left| \frac{a+b+c}{3} \right| \cdot |u-v\omega|$$

$$\leq \frac{2}{3}\sqrt{\mu} \cdot \sqrt{\mu} = \frac{2}{3}\mu$$

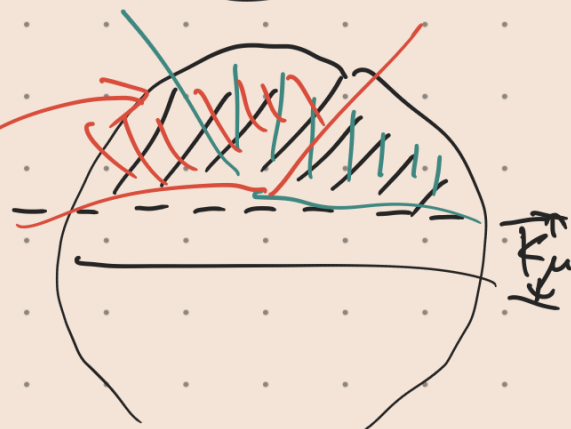
Six cross edges.

$$\Rightarrow (a,u), (a,v), (b,u), (b,v)$$

$$(c,u), (c,v)$$



$$\Rightarrow \left(\frac{a+b+c}{3}, \underline{u-v\omega} \right) \in$$



$$K\mu \leq \left| \left(\frac{a+b+c}{3}, u-vw \right) \right| \stackrel{(*)}{\leq} \frac{2}{3}\mu \quad \hookrightarrow$$
